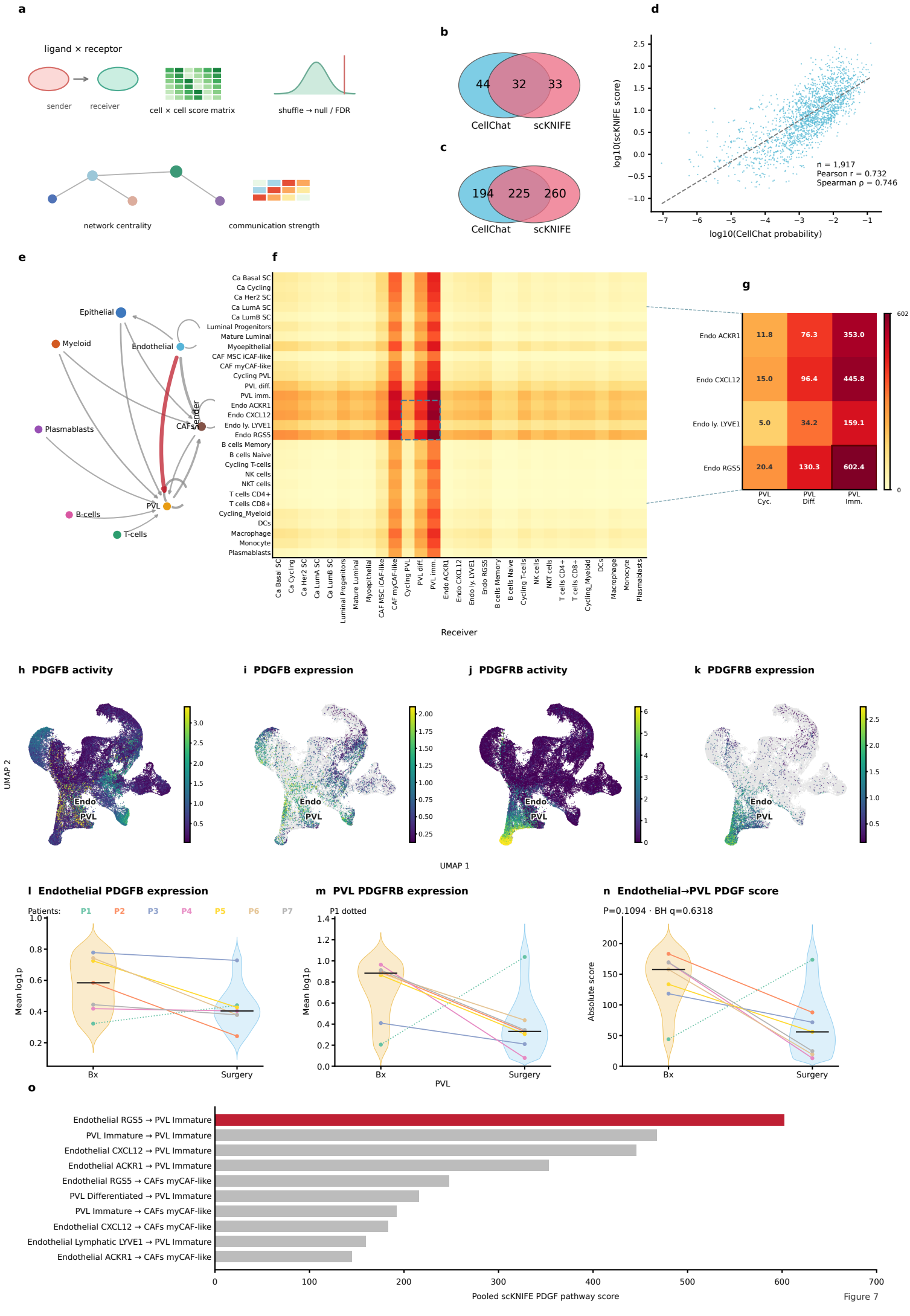

a
ligand × receptor
sender
receiver
cell × cell score matrix
shuffle → null / FDR
network centrality
communication strength
b
44
32
33
CellChat
scKNIFE
c
194
225
260
CellChat
scKNIFE
d
log10(scKNIFE score)
log10(CellChat probability)
n = 1,917
Pearson r = 0.732
Spearman ρ = 0.746
e
Epithelial
Myeloid
Endothelial
Plasmablasts
CAFs
PVL
B-cells
T-cells
Sender
f
Ca Basal SC
Ca Cycling
Ca Her2 SC
Ca LumA SC
Ca LumB SC
Luminal Progenitors
Mature Luminal
Myoepithelial
CAF MSC iCAF-like
CAF myCAF-like
Cycling PVL
PVL diff.
PVL imm.
Endo ACKR1
Endo CXCL12
Endo ly. LYVE1
Endo RGS5
B cells Memory
B cells Naive
Cycling T-cells
NK cells
NKT cells
T cells CD4+
T cells CD8+
Cycling_Myeloid
DCs
Macrophage
Monocyte
Plasmablasts
Receiver
g
Endo ACKR1
Endo CXCL12
Endo ly. LYVE1
Endo RGS5
11.8
76.3
353.0
15.0
96.4
445.8
5.0
34.2
159.1
20.4
130.3
602.4
PVL Cyc.
PVL Diff.
PVL Imm.
602
0
h PDGFB activity
i PDGFB expression
j PDGFRB activity
k PDGFRB expression
UMAP 2
UMAP 1
Endo
PVL
l Endothelial PDGFB expression
Patients:
P1
P2
P3
P4
P5
P6
P7
Mean log1p
Bx
Surgery
m PVL PDGFRB expression
P1 dotted
Mean log1p
Bx
Surgery
PVL
n Endothelial→PVL PDGF score
P=0.1094 · BH q=0.6318
Absolute score
Bx
Surgery
o
Endothelial RGS5 → PVL Immature
PVL Immature → PVL Immature
Endothelial CXCL12 → PVL Immature
Endothelial ACKR1 → PVL Immature
Endothelial RGS5 → CAFs myCAF-like
PVL Differentiated → PVL Immature
PVL Immature → CAFs myCAF-like
Endothelial CXCL12 → CAFs myCAF-like
Endothelial Lymphatic LYVE1 → PVL Immature
Endothelial ACKR1 → CAFs myCAF-like
Pooled scKNIFE PDGF pathway score
Figure 7